

Peak Calculations

Centroid / Peak Center P :

Normal:

The peak center, P , is calculated after subtracting the background as a weighted average of channels under the peak.

$$P = \frac{\sum iC_i}{\sum C_i},$$

where C_i equals the counts in channel i after background subtraction is performed.

Median:

The channel with equal counts to the left and right of it is displayed. This is often used for airborne particle sizing analyzers and aerosol research.

FWHM is calculated as the number of channels in which the counts exceed half of the maximum counts in the peak, with background subtraction corrected linearly for both edges of the peak. When the smooth option is selected, the data in the ROI are first smoothed and then applied to the algorithm above.

Net Area S :

$$S = G - \frac{Bl + Br}{2n} N$$

Uncertainty:

The Net Uncertainty (%) of the Peak defined by:

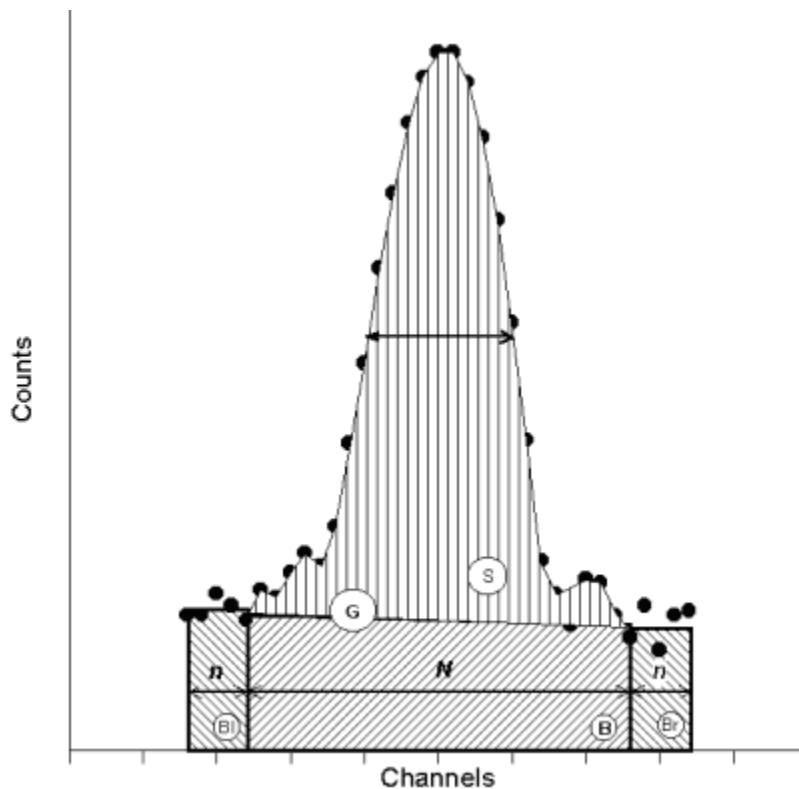
$$\sigma_s = \sqrt{G + \frac{N^2}{4n^2} (Bl + Br)} / (\text{NET AREA}).$$

Net Rate:

The Net Rate is defined by the Net Area / Live_Time.

Gross Area, G : Sum of total counts under the peak excluding the background points.

Example: If background points = 1 and the ROI is three channels wide the Gross Area will be equal to only the center channel. This is to remain consistent with the same boundaries that the Net Area uses in its calculation.



where

G Gross area is the sum of counts under the marked peak

Bl Background left is the area used to calculate the background left of peak center

Br Background right is the area used to calculate the background right of peak center

n Width of both background fields

N Number of channels under the peak

Gaussian Centroid and FWHM Analysis

The centroid and FWHM of a peak can be calculated by fitting a Gaussian curve to the peak, and then using the centroid and FWHM of this Gaussian curve. The advantage of this is that the Gaussian values will be closer to the true values when analyzing a small amount of statistics. The following Gaussian equation is fit to the peak data using a least squares fit:

$$y(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} e^{-\frac{(x-x_0)^2}{2\sigma^2}} + \text{background}(b_1, b_2)$$

$y(x)$ = counts in channel x

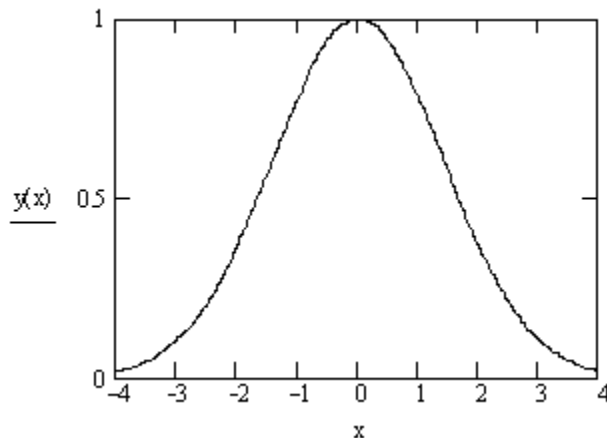
σ = standard deviation

x = channel number

x_0 = centroid

$\text{background}(b_1, b_2)$ = equation of background

The FWHM of the Gaussian curve is equal to 2.35s. Below is a plot of a Gaussian curve with an amplitude of 1, standard deviation of 1, and with no background. The background is just a linear equation which is the same as for the regular analysis.



Variance FWHM Analysis

If the peak being analyzed is known to be Gaussian the FWHM of the curve can be calculated using the **Variance Method**. The Variance Method uses the [Normal Peak Centroid calculation](#) as the mean (μ). The [FWHM of the Gaussian curve](#) is equal to 2.35 times the standard deviation (s) which is the square root of the variance.

The mean μ (peak center), is calculated after subtracting the background as a weighted average of channels under the peak.

$$\mu = \frac{\sum_{ROI} i C_i}{\sum_{ROI} C_i}$$

where C_i equals the counts in channel i after background subtraction is performed.

The variance (σ^2) is calculated by subtracting the mean (Centroid channel) (μ) from the weight (channel i) which is then squared and multiplied by the corrected channel counts (C_i). C_i equals the counts in channel i after background subtraction is performed. This is done for each channel i in the ROI and is then summed. The resulting value is divided by the net area which is the sum of the corrected channel counts C_i . The result is the variance which is the standard deviation squared. The standard deviation is multiplied by 2.35 to get the FWHM.

$$\sigma^2 = \frac{\sum_{ROI} (i - \mu)^2 C_i}{\sum_{ROI} C_i}$$

$$FWHM = 2.35\sigma$$